Khyati Parihar

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EDUCATION

University of California San Diego, La Jolla, California **December 2026**
M.S. Business Analytics (Rady Incoming Scholarship Award Recipient)
Relevant Coursework - Deep Learning & NNs, NLP, Recommender Systems, Large-Scale ML Systems, Causal Inference, Cloud Computing
University of Delhi, New Delhi, India **May 2019**
Bachelor of Arts in Economics (Honors)

SKILLS

ML Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, spaCy, ONNX
Languages & Data: Python, SQL, NoSQL, PostgreSQL, Snowflake, A/B Testing, Prompt Engineering
NLP & LLM: Llama 2/3/3.1, GPT-4, Instruction Tuning, Full Fine-Tuning, Bag-of-Words, Dialogue State Management, Context Engineering, Intent Classification, NLU/NLG, Tableau, Matplotlib
Analytics & Visualization: Tableau, Power BI, AWS QuickSight, Excel, Matplotlib, Google Analytics, Postman API, VBA, Agile, Scrum, Jira, Confluence

WORK EXPERIENCE

AI Consultant, *Netomi, Gurugram, India* **September 2024 – June 2025**

- Fine-tuned **Llama 2, Llama 3, Llama 3.1, and GPT-4** on millions of real-world customer conversations across 40+ enterprise clients using **full fine-tuning and instruction tuning**; achieved **90%+ intent classification accuracy** for both NLU and response generation tasks across production deployments in airlines, retail, telecom, and gaming
- Engineered **multi-intent dialogue optimization** by bifurcating intent clusters and reducing training set size to resolve open-source generation bottlenecks; cut **inference latency from 15s to 6s (60% reduction)** in production NLU pipelines
- Designed and implemented **context engineering frameworks** for LLM behavior control using prompt engineering, Bag-of-Words, and rule-based instruction tuning; evaluated model quality using **precision, recall, F1, deflection rate, and CSAT**, driving continuous accuracy hill-climbing across 40+ client deployments
- Built and curated **large-scale training datasets** from millions of encrypted, masked customer conversations; deployed fine-tuned LLMs to production with **Python-based QA validation pipelines** (pandas, numpy, scikit-learn, spaCy), eliminating manual QA bottlenecks and saving 100+ hours/week
- Designed end-to-end **training data pipelines and schema** from scratch across 40+ clients; drove **accuracy hill-climbing** via systematic error analysis, A/B experiments, and iterative refinement; tracked **precision, recall, and F1** across releases to identify failure patterns and ensure production-ready dataset quality

ML Data Analyst, *Netomi, Gurugram, India* **July 2021 – August 2024**

- Built **intent classification and NLU models from scratch** for individual enterprise clients across airlines, retail, telecom, and gaming; owned full per-client model lifecycle - data ingestion, feature engineering, training, evaluation, and production deployment
- **Optimized LLM parameter footprint** through intent decomposition and training data reduction - splitting overloaded intent clusters into targeted sub-models with fewer parameters; improved response accuracy and deflection rates while reducing per-model resource consumption across client deployments
- Trained and evaluated **early-stage LLM and NLP models** using Python (pandas, numpy, scikit-learn, spaCy); automated response accuracy validation pipelines, eliminating manual QA bottlenecks and delivering **100+ hours/week** in productivity savings
- Utilized SQL to uncover customer patterns, enabling data-driven decisions that boosted CSAT from 80% to more than 95%, and improved adoption/deflection rates

Information Security Analyst, *Coforge Ltd, Greater Noida, India* **November 2019 – June 2021**

- Wrote **complex SQL queries** to extract and analyse security scan data across multiple OpCos; built **weekly Tableau and AWS QuickSight dashboards** tracking ~10 security threats mitigated weekly, accelerating stakeholder decision-making by **30%**
- Designed **Excel-based vulnerability tracking models** with pivot tables and trend analysis to monitor PCI compliance release scans across TrustWave tools (Fusion, AppScanner, TrustKeeper), ensuring **100% regulatory compliance** for secure releases
- Queried and analysed **vulnerability data using SQL** to identify open threat patterns across OpCos; visualised findings in Tableau to prioritise remediation, reducing security exposure by **40%** and accelerating remediation timelines

TECHNICAL PROJECTS

Prediction of House Credit Default Risk, *Independent Analytics Project* **April 2025 – June 2025**

- Analyzed 50,000+ consumer loan applications for deploying ML models and statistical tests, replacing traditional credit-based decisions. Employed diverse forecasting models - Naive Bayes, Gradient Boost, Neural Networks, Decision Trees, and Random Forest, validating model with 88% accuracy, curtailing false positives by **30%** and elevating recall score by **25%**

Applied AI Hackathon - Sydney Airport, *Professional project* **September 2024 – November 2024**

- Designed a risk-mitigative conversational AI model for Sydney Kingsford Smith Airport, through Predictive Analysis along with LLM algorithm and prompt engineering with real-time insights from Google Weather and flight delay data, resulting in **optimized traffic flow, decreased congestion risks, and enhanced passenger experience**

Data Science Hackathon - United Airlines, *Professional project* **March 2024 – August 2024**

- Built the 1st LLM chatbot for United Airlines, performed multiple rounds of iterations for checking classifier's accuracy and response quality, deciding on deployment of various interactive topics. Used SQL and prompts to scrape data from airline's website, contributing to **98%** accuracy in chatbot's performance